

Application Of ULTRA-LOW-POWER CONVERTERS In The IoT

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It is predicted that there will soon be 25 billion permanently-connected sensors to implement the Internet of Things (IoT) in real time. But the challenge is how these devices connected across the globe in different environmental conditions will be powered? In many cases, power from the grid or large batteries to power up these sensors would not be available.

In that case, an ultra-low-power converter to step up the power from ambient energy sources seems to be a perfect solution. An ultra-low-voltage converter can be defined as a converter that can boost very low input voltage autonomously without any external power supply. The ultra-low-voltage converter can step up input voltage as low as 20mV.

Energy harvesting power sources get more interesting for supplying ultra-low power sensor circuits instead of batteries as shown in Fig. 1, which looks like an attractive and free option. However, power management required for each sensor, data-processing units and embedded processors is a critical task, as availability of this free energy is uncertain and very low in power. This article describes a converter circuit that can be used to step up very low power using a self-oscillating circuit.

System block diagram for energy scavenging using an ultra-low-voltage converter is shown in Fig. 2. In this article, each block (of the block diagram) is modelled in LTspice to observe the results in simulation. Detailed description is given below.

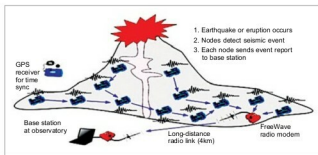


Fig. 1: Energy-harvesting system for the IoT

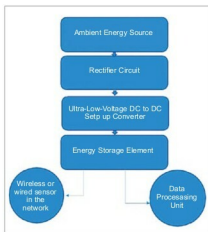


Fig. 2: Block diagram for energy scavenging using ultra-low-voltage converter

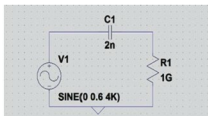


Fig. 3: Ambient energy source model (receiver antenna)

Radio frequency (RF) energy, pressure and temperature are some ambient energy sources. These possess very high internal impedance and, hence, produce output in milli-

volts. This output can be boosted to 3V to 6V range, which could be used to drive low-power sensors and embedded processors.

Energy source. RF waves are emitted by TV or radio-broadcasting antennae. These travel very long distances and so can be used for powering remotely-placed wireless sensors.

Output of these high-frequency, power-carrying RF waves is found to be as low as 600mV. Ambient energy source model is shown in Fig. 3.

AC-to-DC converter. Output from RF antenna is given to bridge rectifier to convert it into constant DC supply voltage. A diode bridge rectifier can be used for rectification purpose here. But output of an AC-to-DC converter would be as low as 600mV, which is not sufficient to turn the diodes that have 700mV forward-voltage drop across these. To reduce forward-voltage drop, we can use MOSFET bridge rectifier circuit, which works perfectly under very-low-voltage input conditions (Fig. 4).

DC-to-DC converter. The converter used here is not a normal step-up converter. It has to step up very low input voltage autonomously. This self-powered converter is based on a self-oscillating circuit like Armstrong oscillator. Advantage of this circuit is ultra-low-input voltage and high-voltage stepping-up capability.

An oscillating circuit is mainly used as a kick-start of a main converter because of the need to skip external-regulated voltage. Most self-oscillating circuits work at mill-watt